

```
import pandas as pd

data = {
    'Weather': ['Sunny', 'Sunny', 'Sunny', 'Rainy', 'Rainy', 'Rainy',
               'Sunny', 'Rainy', 'Sunny', 'Rainy'],
    'Activity': ['Picnic', 'Drink', 'Errands', 'Movie', 'Stay Home',
                'Errands', 'Drink', 'Movie', 'Picnic', 'Stay Home'],
    'Decision': ['Go Out', 'Go Out', 'Go Out', 'Stay In', 'Stay In',
                 'Go Out', 'Go Out', 'Stay In', 'Go Out', 'Stay In']
}

df = pd.DataFrame(data)

# Create CSV file
df.to_csv("weather_decision.csv", index=False)

print("Dataset created successfully!")
print(df)
```

```
Dataset created successfully!
Weather  Activity Decision
0  Sunny    Picnic    Go Out
1  Sunny    Drink    Go Out
2  Sunny    Errands  Go Out
3  Rainy    Movie    Stay In
4  Rainy    Stay Home Stay In
5  Rainy    Errands  Go Out
6  Sunny    Drink    Go Out
7  Rainy    Movie    Stay In
8  Sunny    Picnic    Go Out
9  Rainy    Stay Home Stay In
```

```
data = pd.read_csv("weather_decision.csv")

print("Dataset:")
print(data)
```

```
Dataset:
Weather  Activity Decision
0  Sunny    Picnic    Go Out
1  Sunny    Drink    Go Out
2  Sunny    Errands  Go Out
3  Rainy    Movie    Stay In
4  Rainy    Stay Home Stay In
5  Rainy    Errands  Go Out
6  Sunny    Drink    Go Out
7  Rainy    Movie    Stay In
8  Sunny    Picnic    Go Out
9  Rainy    Stay Home Stay In
```

```
data['Weather_Code'] = data['Weather'].map({
    'Sunny': 1,
    'Rainy': 0
})

data['Activity_Code'] = data['Activity'].map({
    'Picnic': 1,
    'Drink': 2,
    'Errands': 3,
    'Movie': 4,
    'Stay Home': 5
})

data['Decision_Code'] = data['Decision'].map({
    'Go Out': 1,
    'Stay In': 0
})

print(data)
```

	Weather	Activity	Decision	Weather_Code	Activity_Code	Decision_Code
0	Sunny	Picnic	Go Out	1	1	1
1	Sunny	Drink	Go Out	1	2	1
2	Sunny	Errands	Go Out	1	3	1
3	Rainy	Movie	Stay In	0	4	0
4	Rainy	Stay Home	Stay In	0	5	0
5	Rainy	Errands	Go Out	0	3	1
6	Sunny	Drink	Go Out	1	2	1
7	Rainy	Movie	Stay In	0	4	0

8	Sunny	Picnic	Go Out	1	1	1
9	Rainy	Stay Home	Stay In	0	5	0

```
X = data[['Weather_Code', 'Activity_Code']]
y = data['Decision_Code']

print("X:")
print(X)

print("\nY:")
print(y)
```

```
X:
  Weather_Code  Activity_Code
0             1             1
1             1             2
2             1             3
3             0             4
4             0             5
5             0             3
6             1             2
7             0             4
8             1             1
9             0             5

Y:
0     1
1     1
2     1
3     0
4     0
5     1
6     1
7     0
8     1
9     0
Name: Decision_Code, dtype: int64
```

```
from sklearn.model_selection import train_test_split

x_train, x_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=5
)

print("X_train:")
print(x_train)

print("\nX_test:")
print(x_test)

print("\nY_train:")
print(y_train)

print("\nY_test:")
print(y_test)
```

```
X_train:
  Weather_Code  Activity_Code
2             1             3
4             0             5
7             0             4
1             1             2
0             1             1
8             1             1
6             1             2
3             0             4

X_test:
  Weather_Code  Activity_Code
9             0             5
5             0             3

Y_train:
2     1
4     0
7     0
1     1
0     1
8     1
6     1
3     0
Name: Decision_Code, dtype: int64
```

```
Y_test:
9      0
5      1
Name: Decision_Code, dtype: int64
```

```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(
    criterion='entropy',
    random_state=5
)

model.fit(x_train, y_train)

print("Decision Tree model trained successfully!")
```

Decision Tree model trained successfully!

```
y_pred = model.predict(x_test)

print("Predicted Values:")
print(y_pred)

print("\nActual Values:")
print(y_test.values)
```

Predicted Values:
[0 1]

Actual Values:
[0 1]

```
from sklearn import metrics

conf_mat = metrics.confusion_matrix(y_test, y_pred)

print("Confusion Matrix :")
print(conf_mat)
```

Confusion Matrix :

```
[[1 0]
 [0 1]]
```

```
accuracy_score = metrics.accuracy_score(y_test, y_pred)

print("Accuracy Score :", accuracy_score)
print("Accuracy in Percentage :", int(accuracy_score * 100), "%")
```

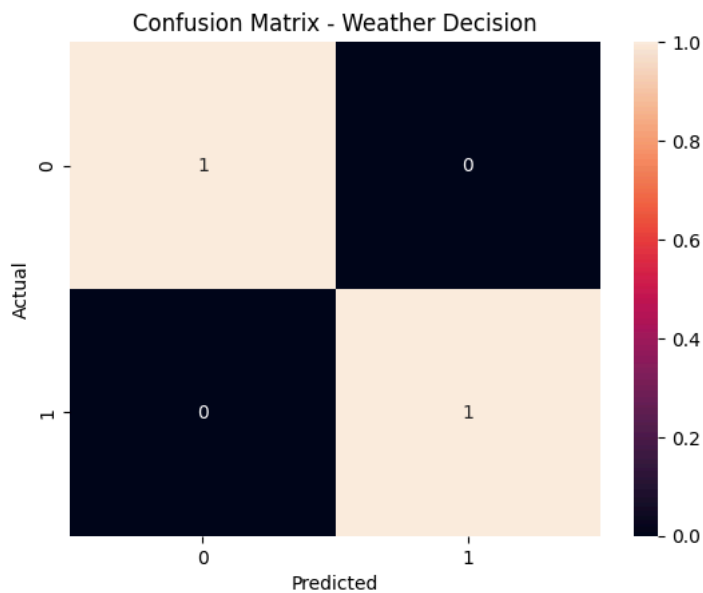
Accuracy Score : 1.0
Accuracy in Percentage : 100 %

```
import seaborn as sn
import matplotlib.pyplot as plt

conf_mat = pd.crosstab(
    y_test,
    y_pred,
    rownames=['Actual'],
    colnames=['Predicted']
)

sn.heatmap(conf_mat, annot=True)

plt.title("Confusion Matrix - Weather Decision")
plt.show()
```

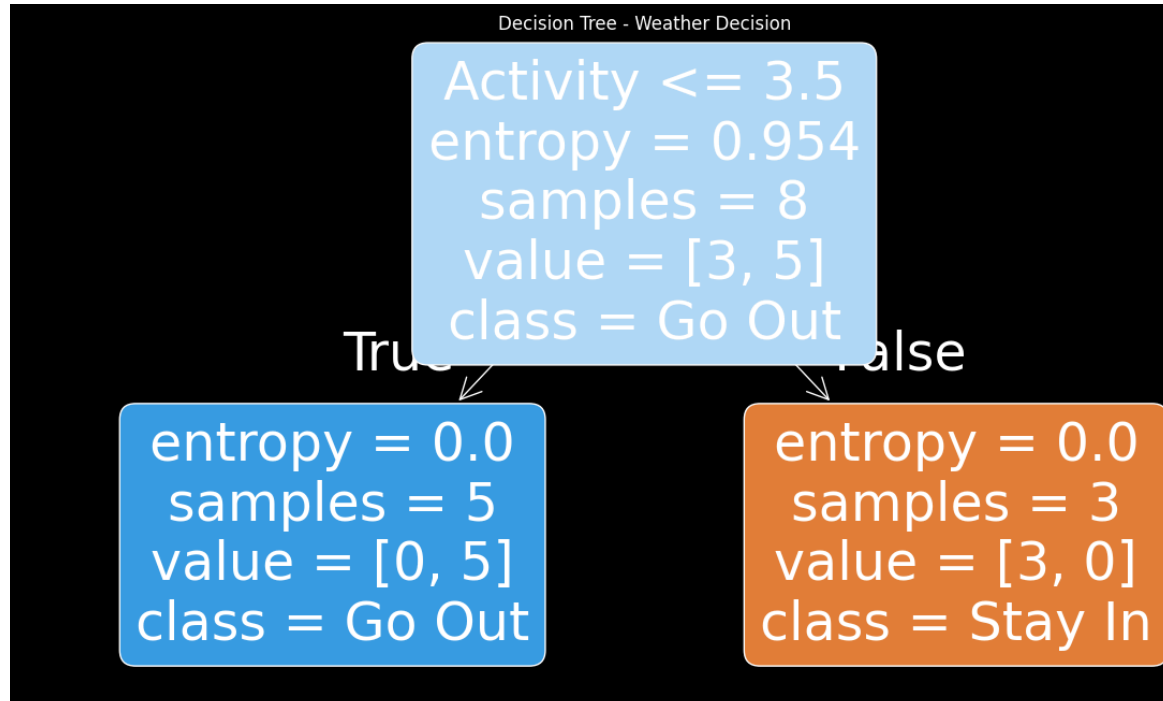


```
from sklearn import tree
import matplotlib.pyplot as plt

plt.figure(figsize=(15, 8))

tree.plot_tree(
    model,
    feature_names=['Weather', 'Activity'],
    class_names=['Stay In', 'Go Out'],
    filled=True,
    rounded=True
)

plt.title("Decision Tree - Weather Decision")
plt.show()
```



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